Practical - 4

Practical Title: JavaScript DOM Manipulation, Event Handling, and UI Interactivity

Problem Definition: Add dynamic UI components such as collapsible FAQ, modal popup, image/content slider, notification banner, hamburger menu, and light/dark theme switcher using JavaScript DOM manipulation and localStorage.

A. Theory & Analysis

1. How are DOM elements selected and modified?

I selected DOM elements using document.getElementById() for unique elements and document.querySelector() / querySelectorAll() for CSS classes. Elements are updated by toggling classes with classList.toggle(), modifying content using textContent or innerHTML, and changing attributes such as data-bs-theme and aria-expanded.

2. Are event listeners attached correctly?

Yes. Event listeners are bound using addEventListener() for click, submit, and input events. Code execution is wrapped in document.addEventListener('DOMContentLoaded', ...) to ensure elements exist before script execution, preventing null reference errors.

3. Is localStorage used for remembering theme choice?

Yes. When the user clicks the theme toggle button, the selected state is saved using localStorage.setItem('theme', 'dark'). When any page loads, an initialization script reads localStorage.getItem('theme') and applies data-bs-theme='dark' to the body, preserving the user's choice across page navigation.

4. Does interactivity improve usability without breaking accessibility?

Yes. The collapsible FAQ updates aria-expanded states for screen readers; the modal popup supports keyboard Escape-key dismissal; the notification banner can be closed easily; and the theme switcher maintains readable text contrast in both light and dark modes.

B. Implementation Details

I implemented interactive JavaScript features across the portal: (1) Collapsible FAQ accordion in pages/about.html; (2) Quick Info modal dialog in pages/dashboard.html; (3) Academic Highlights content slider in js/dashboard.js; (4) Dismissible notice alert; (5) Mobile hamburger menu; and (6) Global persistent light/dark theme switcher in js/common.js saving user preference to localStorage.

C. Evidence of Work

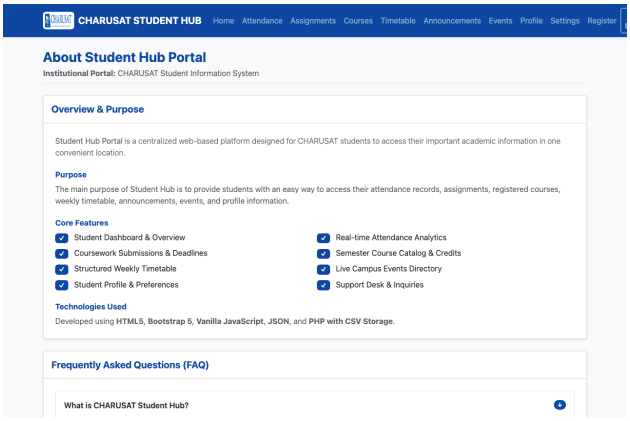


Figure 1: Interactive Collapsible FAQ Accordion with Dynamic Icon Toggling

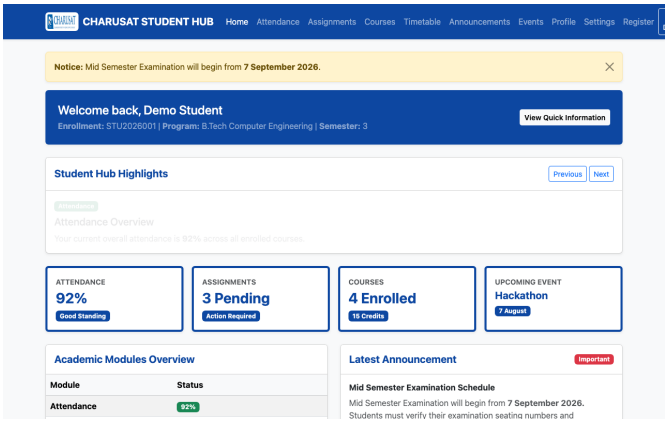


Figure 2: Dashboard with Interactive Carousel Slider and Dismissible Banner

D. Repository Link

GitHub Repository: https://github.com/neel92654/WDF_A_ARP

E. Learning Outcomes

I learned how to manipulate DOM elements, handle user interaction events, create dynamic modals and sliders, and persist client preferences across pages using browser localStorage.